Chapter 17, Problem 1RQ

Problem

Which of the following cannot be a Fourier series?

(a) $t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \frac{t^5}{5}$

(b) 5 sin t + 3 sin 2t − 2 sin 3t + sin 4t

(c) sin t − 2 cos 3t + 4 sin 4t + cos 4t

(d) sin t + 3 sin 2.7t − cos πt + 2 tan πt

(e) $1 + e^{-j\pi t} + \frac{e^{-j2\pi t}}{2} + \frac{e^{-j3\pi t}}{3}$

Step-by-step solution

Step 1 of 5

(A) $t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \frac{t^5}{5}$

$\boxed{\text{Can not be a fourier series}}$

Step 2 of 5

(B) $5\sin t + 3\sin 2t - 2\sin 3t + \sin 4t$

$\boxed{\text{Can be a fourier series}}$

Step 3 of 5

(C) $\sin t - 2\cos 3t + 4\sin 4t + \cos 4t$

$\boxed{\text{Can be a fourier series}}$

Step 4 of 5

(D) $\sin t - 3\sin 2.7t - \cos \pi t + \tan \pi t$

$\boxed{\text{Can be a fourier series}}$

Step 5 of 5

(E) $1 + e^{-j\pi t} + \frac{e^{-j2\pi t}}{2} + \frac{e^{-j3\pi t}}{3}$

$\boxed{\text{Can be a fourier series}}$